TTT4181 - Assignment 2

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1 Introduction

Acoustic waves propagating in confined environments are affected by boundary conditions, giving rise to phenomena such as reflection, resonance and standing waves. Numerical simulations provide a means of investigating these effects. The Transmission Line Matrix (TLM) method is a time-domain numerical technique originally developed for electromagnetic wave propagation and later adapted to acoustics [1]. It represents wave propagation and interactions through a network of interconnected nodes.

In this work, a two-dimensional TLM model of a rigid pipe is implemented to investigate resonant behaviour, standing wave patterns and boundary reflections. A dissipative fluid extension is also introduced to examine the reduction of reflections at the pipe termination. The numerical results are compared with theoretical predictions.

2 Theory

2.1 Acoustic wave propagation and resonance

At 20°C, the speed of sound in air is approximately $c = 343\,\mathrm{m/s}$ [2, Sec. 10.3.2]. Frequency and wavelength are related by [2, Chap. 10]:

$$c = f\lambda. \tag{1}$$

The wavelength relative to the pipe dimensions determines the possible eigenfrequencies and mode shapes. Furthermore, for a rectangular pipe with rigid walls, the ideal theoretical eigenfrequencies are given by

$$f_{mn} = \frac{c}{2}\sqrt{\left(\frac{m}{L}\right)^2 + \left(\frac{n}{W}\right)^2} \tag{2}$$

where L and W denote the length and width of the pipe, respectively. The non-negative integers m and n denote the longitudinal and transverse mode orders, respectively [2, Chap. 13]. The $(1, 0)$ and $(0, 1)$ modes represent the fundamental longitudinal and transverse modes.

Reflections cause incident and reflected waves to interfere, producing standing-wave patterns with spatial pressure maxima and minima. At resonance, the response associated with an eigenmode becomes pronounced.

2.2 TLM method

The Transmission Line Matrix (TLM) method represents a continuous acoustic medium as a discrete network of interconnected nodes. Each node is connected to its neighbouring nodes through transmission lines, along which pressure waves propagate.

The method consists of two main operations:

- Propagation: Pressure waves travel along transmission lines between nodes over discrete time steps.
- Scattering: Incident waves reaching a node are redistributed into outgoing waves along the connected transmission lines.

Through repeated propagation and scattering, the method simulates the time-dependent acoustic pressure field within the discretized domain. Further details on the TLM method are provided in [1].

2.3 Reflection coefficient and boundary conditions

The pressure reflection coefficient r describes the ratio between the reflected and incident pressure waves:

$$r = \frac{p_{\text{ref}}}{p_{\text{inc}}} \tag{3}$$

For harmonic waves, r is complex, where $|r|$ describes the relative reflected pressure amplitude and the phase describes the phase shift [2, Chap. 11].

In the TLM implementation, R denotes the branch reflection coefficient, which is related to the network reflection coefficient G through the relation given by Kagawa et al. [1]:

$$G = \frac{\sqrt{2}(1+R) - (1-R)}{\sqrt{2}(1+R) + (1-R)} \tag{4}$$

In this implementation, the pressure reflection magnitude obtained from the simulated field corresponds to $|G|$. Thus, R and $|r|$ are not generally equal.

At an ideally rigid boundary, the normal particle velocity is zero, resulting in complete pressure reflection ($|r| = 1$). At an absorbing boundary, the reflected pressure amplitude is reduced ($|r| < 1$) [2, Chap. 11].

2.4 Standing Wave Ratio

For a superposition of incident and reflected harmonic waves, the maximum and minimum pressure amplitudes are given by:

$$|p_{\max}| = |p_i|(1 + |r|) \tag{5}$$
$$|p_{\min}| = |p_i|(1 - |r|) \tag{6}$$

where $|p_i|$ is the incident pressure amplitude and $|r|$ is the magnitude of the pressure reflection coefficient.

The standing wave ratio (SWR) is defined as:

$$s = \frac{|p_{\max}|}{|p_{\min}|} \tag{7}$$

Combining the expressions above gives:

$$s = \frac{1 + |r|}{1 - |r|} \tag{8}$$

Rearranging yields the relationship used to estimate the pressure reflection magnitude from the simulated standing-wave pattern [2, Sec. 10.6]. In the present TLM implementation, this corresponds to the magnitude of the network reflection coefficient G:

$$\boxed{|r| = \frac{s-1}{s+1} = |G|} \tag{9}$$

2.5 Dissipative termination

Following Kagawa et al. [1, Sec. 2.4], a 1 m dissipative extension is added to the original pipe to reduce reflections at the termination. The outermost boundary remains rigid, while distributed losses attenuate waves propagating through the extension.

The losses are controlled by a parameter z, which modifies the node pressure during scattering:

$$p_{\mathrm{node}} = \frac{2\sum_{n=1}^{4} P_n}{4+z} \tag{10}$$

where P_n denotes the incident pressure pulses at the node. Setting $z = 0$ recovers the lossless relation. Constant, linear and quadratic spatial loss profiles are investigated.

3 Methodology

3.1 Numerical model and discretization

The simulated domain represents a rectangular air-filled pipe with a length of 2 m and a width of 0.2 m. The air temperature is set to 20°C, corresponding to a sound speed of $c_0 = 343\,\mathrm{m/s}$ and a density of $\rho_0 = 1.204\,\mathrm{kg/m^3}$.

The domain is discretized into cell-centred TLM nodes with a uniform spacing of $\Delta x = 0.01\,\mathrm{m}$, giving a grid of 200×20 nodes. At the maximum frequency of 2000 Hz, the minimum wavelength is $\lambda_{\min} = c_0/f_{\max} = 0.1715\,\mathrm{m}$, corresponding to approximately 17 mesh points per wavelength.

The time step is determined by the propagation scheme:

$$\Delta t = \frac{\Delta x}{\sqrt{2}c_0}. \tag{11}$$

The walls are located half a cell outside the outermost nodes to preserve the specified physical dimensions. The 2D TLM model was implemented in Python.

3.2 Sources, boundaries and measurements

The excitation is introduced as a soft source by adding the source signal to an incoming pressure pulse. Depending on the task, the excitation consists of a sinusoidal sweep, a single-timestep impulse or a harmonic signal.

Unless otherwise specified, the pipe ends and sidewalls have a branch reflection coefficient of $R = 1$. The reflection coefficient at the pipe ends is reduced to $R = 0.1$ or $R = 0.3$ in the corresponding investigations. The pressure field is recorded at selected locations or averaged across the pipe width.

For standing-wave measurements, the width-averaged pressure amplitude is estimated by fitting a sinusoidal function over the final ten periods. The two minima closest to the right boundary and the maximum between them are identified. The first minimum is used to calculate the standing wave ratio.

3.3 Simulation procedure

The simulation settings for the individual tasks are summarized in Table 1.

Table 1: Overview of the simulation setups.

Task	Simulation setup
1	Sine sweep from 100 to 2000 Hz
2	Single-timestep impulse excitation
3	Harmonic point source; investigation of the $(1, 0)$ mode
4	Reduced end reflection, $R = 0.1$
5	Harmonic excitation at 200 and 500 Hz; SWR measurements
6	SWR measurements for $R = 1$ and $R = 0.3$
7	1 m dissipative extension with varying loss profiles

The sine sweep was applied for 2 s, followed by a 0.3 s ring-down period, while the impulse response was simulated for 1 s. The sweep parameters were adjusted slightly to ensure that the signal started and ended at a zero crossing. Task 3 used 30 periods of harmonic excitation with a cosine fade-in over the first three periods. For tasks 5–7, the harmonic simulations ran for 0.5 s, with a cosine fade-in over the first 30% of the simulation time.

For Task 7, the pipe was extended by 1 m, with distributed loss introduced in the extension and a rigid boundary at its far end. The reflection magnitude was estimated from the standing-wave ratio in the original 2 m section. The width-averaged pressure amplitude was obtained by fitting a sinusoid over the final ten periods at each position. The two minima closest to the right boundary and the maximum between them were identified, and the first minimum was used to calculate the SWR and reflection magnitude.

The spatial loss profiles in the extension ($L \leq x \leq L + L_{\text{ext}}$) were defined using the normalized coordinate

$$\xi = \frac{x - L}{L_{\text{ext}}}, \qquad 0 \leq \xi \leq 1,$$

where $L = 2\,\text{m}$ and $L_{\text{ext}} = 1\,\text{m}$. The profiles were defined as

$$z(x) = \begin{cases} z_{\max}, & \text{constant,} \\ z_{\max}\xi, & \text{linear,} \\ z_{\max}\xi^2, & \text{quadratic.} \end{cases}$$

The maximum loss parameter was varied over $z_{\max} \in \{0.02, 0.05, 0.08, 0.10, 0.15, 0.20, 0.30, 0.50, 1.00\}$.

4 Results

This section presents the simulation results for the described methodology. The results are compared with theoretical predictions where applicable. Supporting figures are provided in Appendix A, as the requirement of 6 pages could not be met otherwise.

4.1 Linear sine sweep

A linear sine sweep from 100 Hz to 2000 Hz was applied at the lower left corner of the pipe, and the pressure response was recorded at the lower right corner. Resonance frequencies were identified from the resulting frequency spectrum (Figure A.1) and compared with theoretical values based on Equation 2, as shown in Table 2.

Table 2: Comparison of selected theoretical and simulated resonance frequencies.

Mode	Theory [Hz]	Simulation [Hz]
$(2, 0)$	171.50	171.3
$(4, 0)$	343.00	343.0
$(6, 0)$	514.50	514.3
$(8, 0)$	686.00	685.6
$(10, 0), (0, 1)$	857.50	857.0
$(11, 2)$	1957.28	1955.6

The fundamental longitudinal mode, $(1, 0)$, occurs at 85.75 Hz and is therefore below the sweep's starting frequency of 100 Hz. The first transverse-mode cut-on frequency is 857.5 Hz. The simulated resonance frequencies corresponds well with theoretical predictions.

4.2 Dirac pulse excitation

The sine sweep was replaced by a single-timestep Dirac pulse, while keeping the source and receiver positions unchanged. The resulting frequency spectrum is shown in Figure A.2.

The identified resonance frequencies closely match those obtained using the linear sine sweep. The Dirac pulse excites a broad range of frequencies simultaneously, allowing multiple eigenmodes to be identified from a single simulation.

4.3 Harmonic excitation at the (1,0) mode

A harmonic point source was placed at the lower left corner and operated at 85.8 Hz, corresponding to the $(1, 0)$ mode. The pressure distribution along the pipe centreline was recorded at four instants within the final period, after 29 periods of excitation (Figure A.3).

The resulting pressure distribution exhibits a node approximately at the midpoint of the pipe, consistent with the theoretical mode shape. The pressure minimum is not exactly zero, and minor deviations from the ideal mode shape are observed. These deviations may be attributed to the finite spatial discretization and the numerical nature of the simulation.

4.4 Reduced reflection coefficient

The simulation setup was kept identical to that of Task 1, except that the branch reflection coefficient at both pipe ends was reduced from $R = 1.0$ to $R = 0.1$. The resulting pressure spectrum is shown in Figure A.4.

Compared to the fully reflective case ($R = 1.0$), the reduced reflection coefficient results in less pronounced resonance peaks. This indicates weaker standing waves due to increased energy dissipation at the boundaries.

4.5 Standing wave ratio and reflection coefficient

The standing wave ratio (SWR) was measured at 200 Hz and 500 Hz for reflection parameters $R = 1.0$ and $R = 0.3$. The two pressure minima closest to the right boundary and the maximum between them were identified. The pressure reflection magnitudes were then calculated using Equation 9. The results are presented in Table 3.

Table 3: Measured pressure extrema, standing wave ratios and pressure reflection magnitudes. Positions are given in parentheses.

R	G	f [Hz]	$\lvert p_{\min,1}\rvert$ (x [m])	$\lvert p_{\min,2}\rvert$ (x [m])	$\lvert p_{\max}\rvert$ (x [m])	s	$\lvert r\rvert$
1.00	1.000	200	$2.118 \cdot 10^{-5}$ (1.571)	$2.625 \cdot 10^{-5}$ (0.714)	0.04094 (1.142)	1933.186	0.999
1.00	1.000	500	$2.133 \cdot 10^{-4}$ (1.829)	$8.958 \cdot 10^{-5}$ (1.486)	0.07305 (1.657)	342.541	0.994
0.30	0.448	200	0.01951 (1.571)	0.01951 (0.714)	0.05125 (1.143)	2.627	0.449
0.30	0.448	500	0.02367 (1.829)	0.02367 (1.486)	0.06219 (1.657)	2.628	0.449

For $R = 1.0$, the measured pressure reflection magnitude was close to unity at both frequencies. For $R = 0.3$, the measured value was approximately $|r| = 0.449$, closely matching the theoretical network reflection coefficient $G = 0.448$ from Equation 4. This confirms that the TLM parameter R is not generally equal to the effective pressure reflection magnitude $|r|$.

4.6 Dissipative termination

A 1 m dissipative fluid extension was added to the right end of the pipe, with a rigid boundary at its far end. Constant, linear and quadratic loss profiles were tested for different values of $z_{\max}$. The lowest measured reflection magnitudes are presented in Table 4.

Table 4: Lowest measured pressure reflection magnitudes for the 1 m lossy extension.

f [Hz]	Loss profile	$z_{\max}$	$\lvert r\rvert$
200	Linear	0.10	0.0834
1000	Quadratic	0.50	0.0007

The results show that the most effective loss profile depends on frequency and the spatial distribution of dissipation. The extension achieved a particularly low reflection magnitude at 1000 Hz, while a larger residual reflection remained at 200 Hz.

At 1000 Hz, the frequency exceeds the first transverse-mode cutoff frequency. The width-averaged pressure isolates the plane-wave component, while higher-order transverse modes may also contribute to the total reflected field.

5 Discussion

The simulated resonance frequencies mostly agreed with the theoretical predictions. Both the linear sine sweep and the Dirac pulse identified the same resonant modes, demonstrating that both excitation methods are suitable for identifying the pipe's eigenfrequencies. The harmonic excitation at the $(1, 0)$ mode also produced the expected longitudinal mode shape, with a pressure node near the midpoint of the pipe.

Reducing the reflection coefficient from $R = 1.0$ to $R = 0.1$ resulted in less pronounced resonance peaks. This is consistent with increased energy dissipation at the boundaries, which weakens the standing waves inside the pipe.

The SWR measurements showed that the effective pressure reflection magnitude $|r|$ is not generally equal to the TLM reflection parameter R. For $R = 0.3$, the measured value agreed closely with the theoretical value G obtained from Equation 4. For $R = 1.0$, the reflection magnitude was close to unity, as expected for a fully reflective boundary. It should be noted that, for $R = 1.0$, the very high SWR values are finite-time estimates, as the lossless system does not reach a true steady state.

Finally, the dissipative extension reduced reflections at both investigated frequencies. However, the most effective loss profile differed between 200 Hz and 1000 Hz, indicating that the reflection reduction depends on the frequency and spatial distribution of losses. The extension did not provide a perfectly anechoic termination. At 1000 Hz, this result applies to the width-averaged pressure component, as discussed in Section 4.

6 Conclusion

The implemented 2D TLM model reproduced the main acoustic phenomena in the pipe, including resonances, standing waves and boundary reflections. The simulated resonance frequencies agreed well with theoretical predictions, and the SWR measurements confirmed the relationship between the TLM reflection parameter and the effective pressure reflection magnitude.

The results further demonstrated that reducing boundary reflections weakens resonances, while a dissipative extension can substantially reduce reflections at the termination. Overall, the TLM method proved useful for studying acoustic wave propagation and boundary effects in a confined domain.

References

[1] Y. Kagawa, T. Tsuchiya, B. Fujii, and K. Fujioka, "Discrete huygens' model approach to sound wave propagation," *Journal of Sound and Vibration*, vol. 218, no. 3, pp. 419–444, 1998.

[2] S. L. Garrett, *Understanding Acoustics: An Experimentalist's View of Sound and Vibration*, 2nd ed. Springer, 2020.

A Additional simulation figures

This appendix contains the pressure spectra and spatial pressure distributions from the simulations discussed in Section 4.

A.1 Linear sine sweep (Task 1)

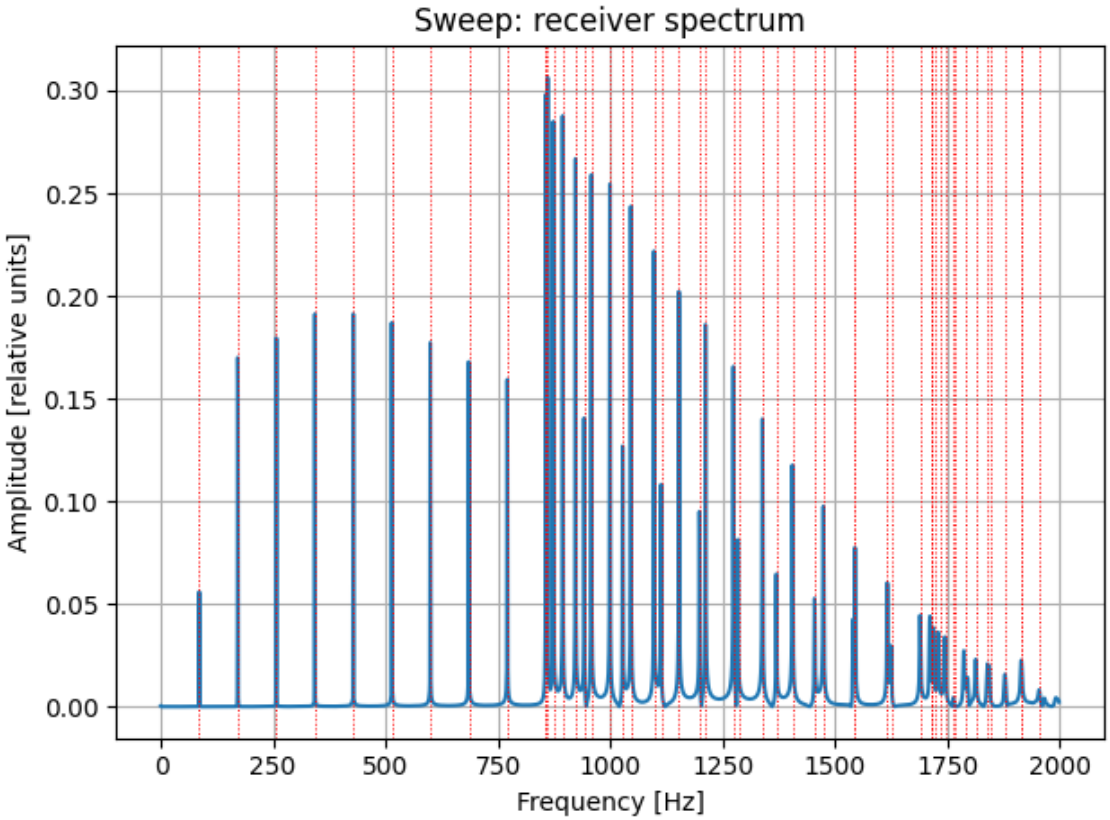


Figure A.1: Frequency spectrum of the pressure response at the receiver for a linear sine sweep excitation. Red vertical lines indicate theoretical resonance frequencies.

A.2 Dirac pulse excitation (Task 2)

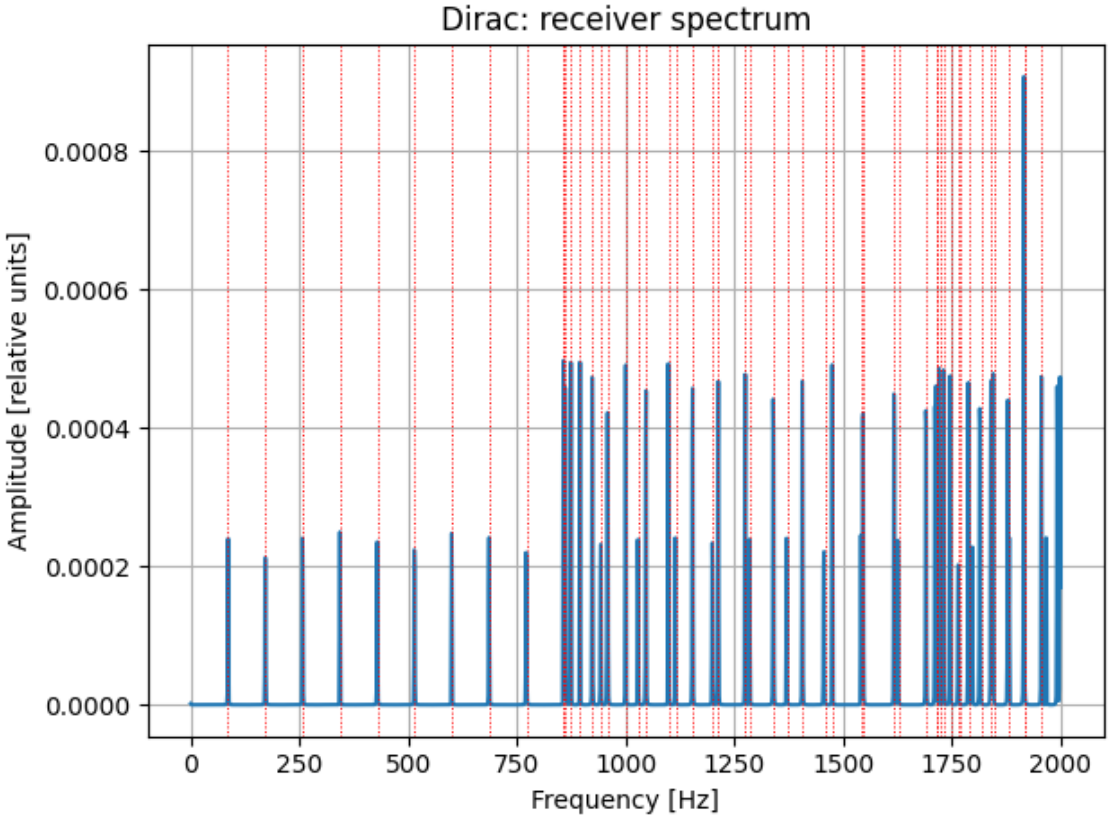


Figure A.2: Frequency spectrum of the pressure response at the receiver for Dirac pulse excitation. Red vertical lines indicate theoretical resonance frequencies.

A.3 Harmonic excitation at the $(1, 0)$ mode (Task 3)

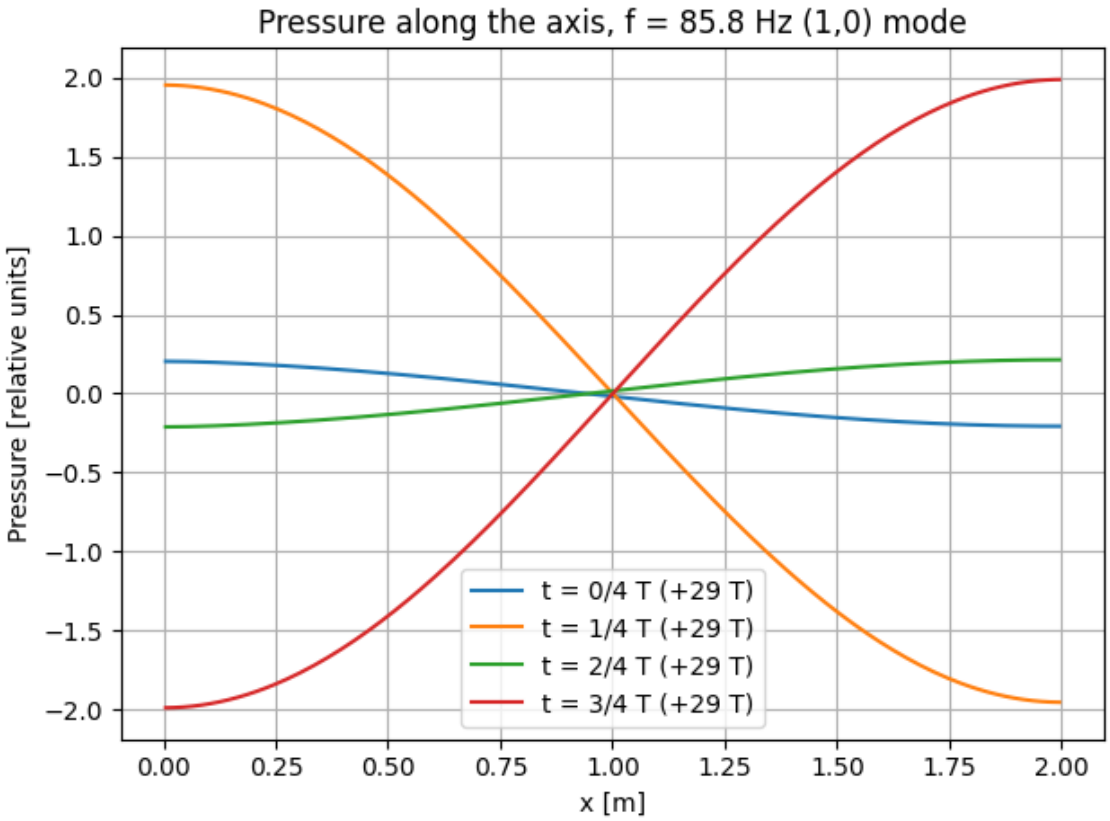


Figure A.3: Pressure distribution along the pipe centreline at four instants within the same period of harmonic excitation at the $(1, 0)$ mode.

A.4 Reduced reflection coefficient (Task 4)

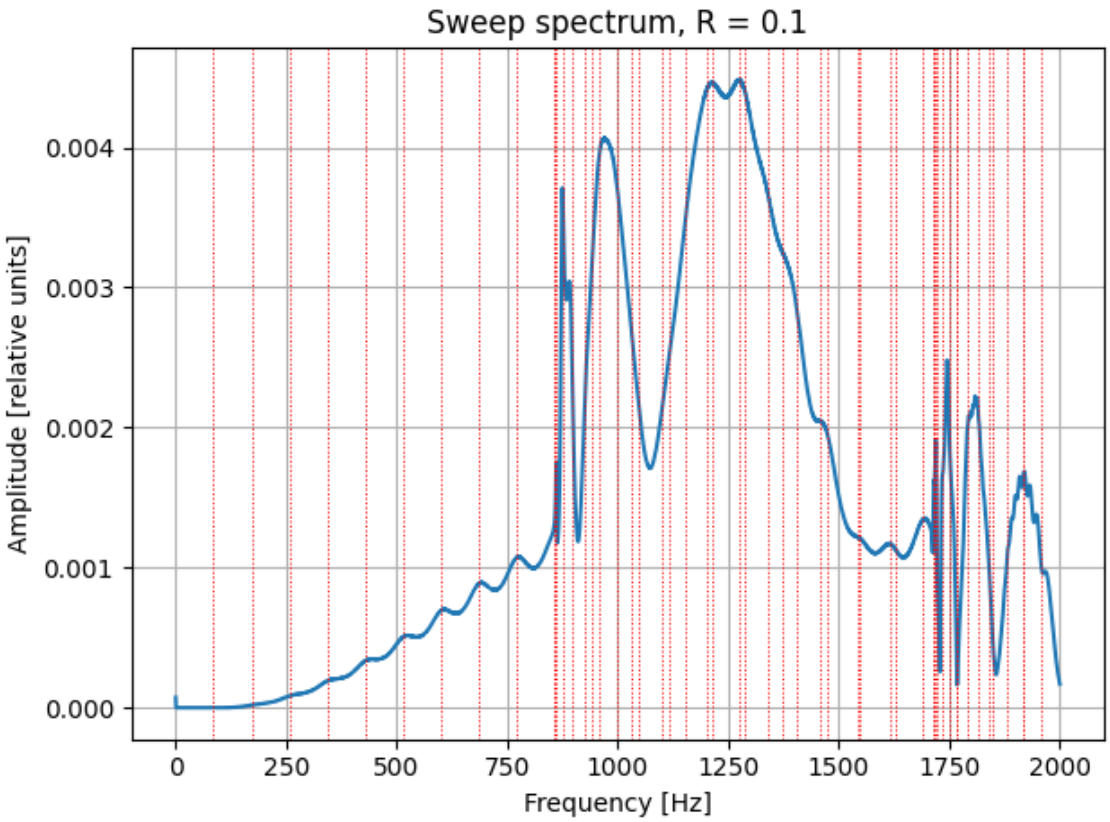


Figure A.4: Frequency spectrum of the pressure response for $R = 0.1$, compared with the theoretical resonance frequencies.

B Declaration on the use of AI

AI was relied on substantially as a tool throughout the work. The utilization of both ChatGPT and Claude AI revolved around the following scenarios:

- Generating code. By supplying Claude AI with both Kagawa [1] and the investigative intentions of the work, a foundation of Python script was built. This code was then modified and verified by hand, and cross referenced through ChatGPT and a test script.
- As a tool of instant feedback when writing the report. A suggested text would be written by the author, whom would then ask ChatGPT for feedback regarding contents, grammar and overall precision. Suggested changes from the AI was adapted when fitting.

C Python code

The results in section 4 was obtained through a Python script (written through the assistance of AI, as aforementioned in Appendix B). The .zip-file containing the Python script is submitted alongside the report. You might also retrieve the code by accessing the following GitHub repo:

`https://github.com/Eek-JensenCEO/TTT4181-Assignment-2.git`